PeriGen

Advanced
Perinatal Systems

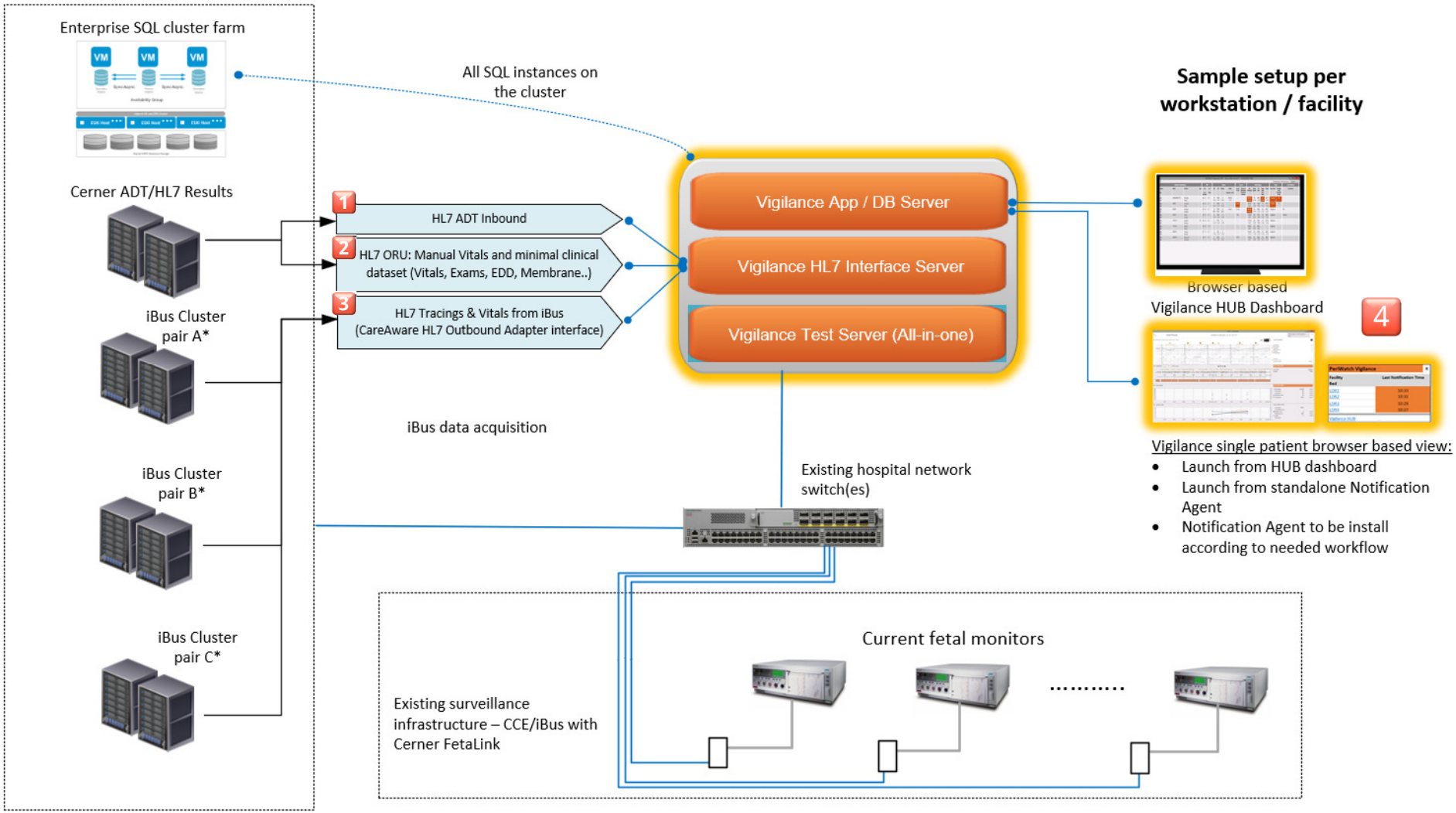
PeriWatch Vigilance

Architecture/Interface Diagrams in Cerner Environment

Jan. 2020 update

Phase II – All Sites: Vigilance Architecture/Data Interfaces with Cerner FetaLink

(All tracing feed Vigilance via HL7 from Cerner CareAware HL7 Outbound Adapter)



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HL7 ADT And Results Interfaces

Standard ADT Inbound Interface: 1️⃣

- Mandatory
- Expectation of a single inbound interface for a multi-facility environment.
- All inbound HL7 interfaces are handled by an included instance of Rhapsody interface engine on the Vigilance server.
- See PeriWatch Vigilance Interface Specifications document.

Standard ORU Result Inbound Interface: 2️⃣

- Mandatory
- Expectation of a single inbound interface to receive the required manually document fields by OB clinicians. These fields are mandatory input to the Vigilance system. These include Manual vitals, Cervical Exams, GA/EDD, Gravida, Parity, # of fetuses etc.
- For a full list of the required fields, refer to section 1.4 of the PeriWatch Vigilance Interface Specification document

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CareAware HL7 Outbound Interface – Tracing Data via HL7

Fetal Tracings Via HL7 Interface: 3️⃣

- Mandatory
- Cerner CareAware receives and manages all device data via iBus. This includes fetal tracing near real-time data collected from fetal monitors via CCE devices. This is the same feed that is used to send tracing data to the FetalLink surveillance system.
- The CareAware infrastructure can expose this data as an HL7 outbound interface data stream. This interface is called the “CareAware HL7 Outbound Interface”
- In is possible that multiple CareAware servers are used at larger enterprises. In order to receive all fetal tracing streams in the enterprise, all servers must provide an HL7 interface to the Vigilance server. Multiple sender with one receiver is supported.
- This interface needs to be purchased and enabled by Cerner and point to the Vigilance Rhapsody inbound TCP port.

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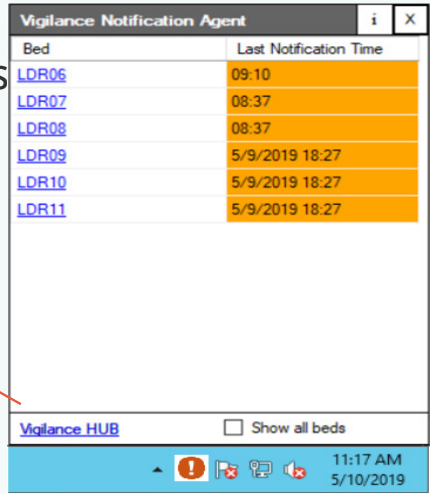
PeriWatch Vigilance Notification Agent

Small application to notify clinicians of Vigilance state change: 4️⃣

- Strongly recommended
- On top of the main Vigilance client which is 100% browser based (not deployment required, just a URL), Vigilance Notification Agent (VNA) is installed on select workstation where clinicians need to see Vigilance notifications (example: bedside stations or nursing stations)
- The VNA needs to be deployed on those clinically significant stations according to the clinical team guidance.
- The VNA is:
  - Independent of any direct Cerner integration at the bedside
  - Deployable by customer IT staff to the designated stations
  - Will pop-up in the foreground with any new Vigilance notifications a small portion of the bottom right of the screen
  - See PeriWatch Vigilance Notification Agent – User Guide for more information

Sample notification agent appearance on a configured workstation



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HL7 Inbound Results Interface

Fields Required For Full Vigilance Utilization

When not present, >=36 WK assumed + Notice

Mandatory

Not mandatory but very useful during clinical review

• EDD/LMP	• Previous Vaginal Delivery
• Gestational Age	• Attempted VBAC
• Gravida	• Number of Fetuses
• Para	• Dilation
• Term	• Effacement
• Pre-Term	• Station
• AB Induced	• Presentation
• AB Spontaneous	• Epidural in Use (DateTime)
• Living	

Defaults to NO

Defaults to NO

Defaults to 1

Mandatory

Effect on Curve

- Maternal Heart Rate
- Maternal BP
- Maternal SpO2
- Respiratory Rate
- Temperature

Required to have a full trend of Maternal vitals to complement those Originating from the monitor

Mandatory

Strongly recommended

Useful for clinicians

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